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Sant Gajanan Maharaj College of Engineering, Mahagaon

Site – Chinchewadi, Tal-Gadhinglaj, Dist-Kolhapur, PIN :- 416503

A Mini- project Report On

“AIR QUALITY INDEX (AQI)”

Under The Guidance of

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(ACADEMIC YEAR 2025-26)



“Sant Gajanan Maharaj College of Engineering, Mahagoan”

SITE –CHINCHEWADI, TAL- GADHINGLAJ, DIST- KOLHAPUR

Certificate

This is to certify that the following students of Six semester of B. Tech in Electronics & Telecommunication Engineering of Institute **SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING, MAHAGON -416503**. has completed

“Air Quality Index” Mini Project for academic year 2025 - 2026 as prescribed in the curriculum.

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Finally, we dedicate this work to our parents. We express our deepest gratitude and appreciation to all our family members, friends, and faculty members, whose constant support and blessings have been a great source of inspiration in completing this work.

Abstract

This project presents the design and implementation of a VLSI-based Air Quality Index (AQI) monitoring system using an FPGA platform. The system utilizes an MQ135 gas sensor to detect air pollutants and generate an analog signal proportional to air quality. This signal is converted into digital form using an SPI-based Analog-to-Digital Converter (MCP3008), enabling efficient data processing within the FPGA.

The FPGA, implemented using Verilog HDL, performs real-time data acquisition, AQI calculation, and classification based on predefined air quality ranges. The processed results are displayed on a 16×2 LCD, providing a simple and user-friendly interface for monitoring air quality conditions. The use of SPI communication ensures reliable and modern data transfer between components.

The proposed system demonstrates a hardware-efficient and real-time solution for air quality monitoring, reducing dependency on software-based processing. The modular design allows easy scalability and provides a strong foundation for future enhancements such as multi-sensor integration, IoT connectivity, and System-on-Chip (SoC) implementation.

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CHAPTER 1

1.1 Introduction

Air pollution has become one of the most critical environmental challenges in modern society, affecting human health, climate, and overall quality of life. Rapid industrialization, urbanization, and increasing vehicular emissions have significantly contributed to the deterioration of air quality. Continuous monitoring and assessment of air pollution levels are therefore essential for ensuring public safety and environmental protection.

The AQI is a standardized metric used to represent the concentration of various pollutants in the air in a simplified and understandable form. Regulatory bodies such as the Central

Pollution Control Board (CPCB) and the United States Environmental Protection Agency (EPA) define AQI categories such as Good, Moderate, Poor, and Hazardous, enabling easy interpretation of air quality conditions. However, most conventional AQI monitoring systems are software-based and rely on microcontrollers or cloud platforms for data processing.

In this project, a hardware-oriented approach is proposed by designing a VLSI-based AQI monitoring system using Verilog HDL implemented on an FPGA platform. The system acquires analog air quality data using gas sensors, converts it into digital form using a Serial Peripheral Interface (SPI)-based Analog-to-Digital Converter (ADC), and processes the data within the FPGA. The AQI value is computed using hardware-efficient techniques and classified according to standard AQI ranges.

The proposed design emphasizes modular architecture, real-time processing capability, and hardware efficiency. By integrating sensor interfacing, SPI communication, AQI computation, and display control within a single FPGA-based system, the project

demonstrates the feasibility of implementing environmental monitoring solutions using VLSI design methodologies. This approach also provides a foundation for future development of System-on-Chip (SoC)-based air quality monitoring systems.

1.2 Problem Statement:

This project aims to develop a hardware-efficient AQI monitoring system using FPGA. The system should accurately acquire air quality data from sensors, process it using digital logic, and classify pollution levels based on standard AQI ranges. The design should be cost-effective, real-time, and suitable for practical environmental monitoring applications.

1.3 Objectives:

1. To Design and implement an FPGA-based system using Verilog HDL to acquire and process air quality data.
 2. To Develop an SPI-based interface to read sensor data through an Analog-to-Digital Converter (ADC).
 3. To Implement an AQI calculation and classification system based on standard air quality ranges.
 4. To Ensure the system is hardware-efficient, real-time, and suitable for practical environmental monitoring applications.
-

CHAPTER -2

2.1 Block diagram:-

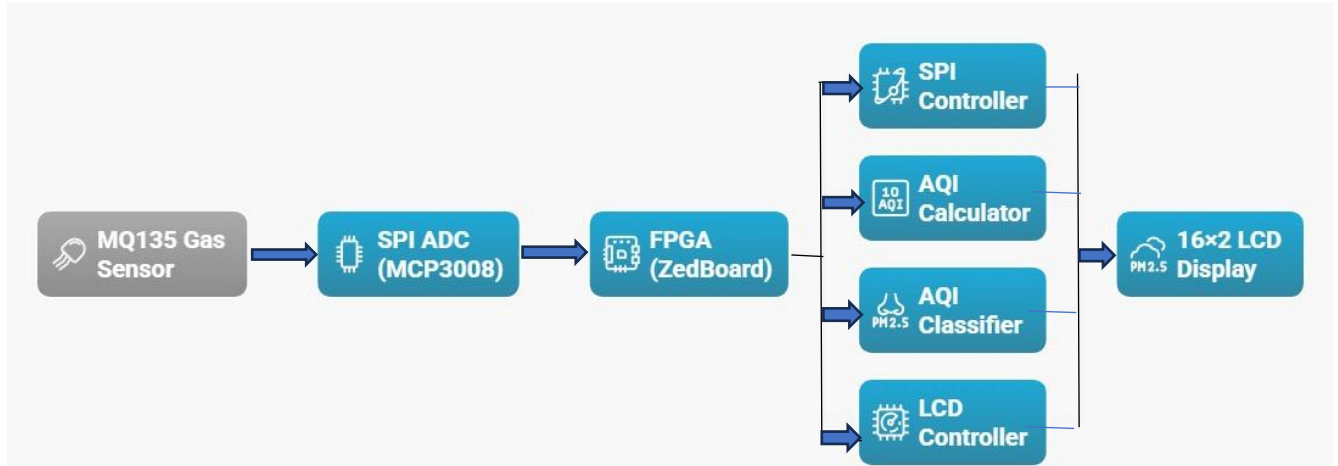


Fig.1 Block diagram of AQI System.

Working: -

As shown in Fig. 1.1, the above block diagram illustrates the system architecture, and the detailed working of each component is explained below

MQ135 Gas Sensor: -

The MQ135 Gas Sensor acts as the initial interface of the system, where air quality is detected. It senses the presence of harmful gases such as CO₂, ammonia, and smoke in the environment. Based on the concentration of these gases, the sensor produces a corresponding analog voltage signal. This signal represents the level of air pollution and is passed to the next stage for further processing.

SPI ADC (MCP3008): -

The SPI-based Analog-to-Digital Converter (ADC) is responsible for converting the analog signal from the gas sensor into a digital format. The MCP3008 reads the sensor's

analog voltage and converts it into a 10-bit digital value. This conversion is essential because the FPGA can only process digital signals. The ADC communicates with the FPGA using the SPI protocol, ensuring efficient and reliable data transfer.

FPGA (Zed Board): -

The FPGA acts as the central processing unit of the system. It receives digital data from the ADC through SPI communication. Inside the FPGA, multiple modules such as the SPI controller, AQI calculator, AQI classifier, and LCD controller are implemented using Verilog HDL. The FPGA processes the sensor data, converts it into an AQI value, and classifies it into categories like Good, Moderate, or Poor based on predefined ranges. This makes the FPGA the “brain” of the system.

AQI Calculator & Classifier: -

The AQI calculator processes the incoming digital data and converts it into a normalized AQI value. The classifier then compares this value with predefined AQI ranges to determine the air quality category. This stage ensures that raw sensor data is translated into meaningful and standardized information that can be easily understood.

LCD Display (16×2):-

The LCD display serves as the output interface of the system. It shows the calculated AQI value along with the corresponding air quality status. The FPGA controls the LCD through digital signals, updating the display in real time. This allows users to easily monitor the current air quality condition in a clear and user-friendly manner.

2.2 Circuit Diagram :-

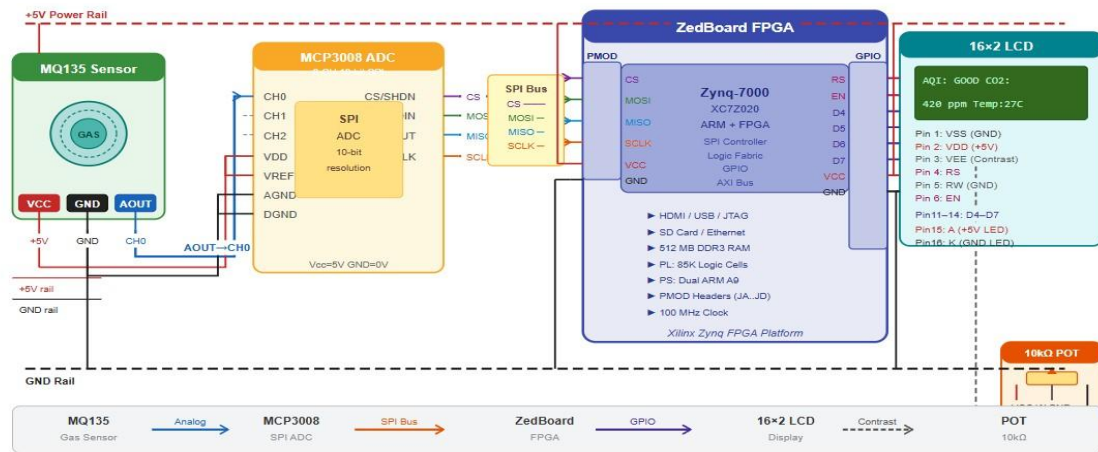


Fig.2 Circuit view AQI System

From Fig. 2 the overall implementation and working of the AQI system can be understood. The detailed explanation of each component is given below.

2.2.1 Zed Board (FPGA Unit)

The core of this circuit is the **Zed Board**, which acts as the intelligent processing unit of the system. It receives digital data from an SPI-based Analog-to-Digital Converter (ADC) and processes it using Verilog-based logic. The FPGA is programmed to control communication, perform AQI calculation, classify air quality levels, and display the results. Hence, it acts as the central “brain” of the system, responsible for all decision making operations.

2.2.2 MQ135 Gas Sensor

The air quality sensing is performed using the **MQ135 gas sensor**, which detects harmful gases such as CO₂, ammonia, and smoke. When exposed to polluted air, the sensor produces a varying analog voltage corresponding to the concentration of these gases.

Since this analog signal cannot be directly processed by the FPGA, it is forwarded to the ADC for further processing.

2.2.3 MCP3008 ADC (SPI Based)

The **MCP3008 ADC** is used to convert the analog signal from the gas sensor into a digital value. This conversion is essential because the FPGA can only process digital signals. The ADC communicates with the FPGA using the Serial Peripheral Interface (SPI) protocol, ensuring accurate and efficient data transfer.

2.2.4 SPI Communication Interface

The communication between the ADC and FPGA is carried out using SPI signals such as **SCLK (clock), MOSI, MISO, and CS (chip select)**. The FPGA generates the clock signal and controls the communication process, while the ADC transmits the converted digital data through the MISO line. This structured communication ensures reliable data exchange between analog and digital domains.

2.2.5 AQI Calculator and Classifier

Inside the FPGA, the received digital data is processed through multiple modules. The SPI controller first reads the data from the ADC, which is then sent to the AQI calculator. The calculator converts the sensor data into a normalized AQI value.

Further, the classifier compares this value with predefined AQI ranges to determine the air quality level, such as **Good, Moderate, or Poor**. These operations are implemented using hardware logic, ensuring fast and real-time processing.

2.2.6 LCD Display (16×2)

The output of the system is displayed on a **16×2 LCD**, which acts as a user-friendly interface. The FPGA controls the LCD using signals such as **RS, EN, and data lines**. The display shows both the AQI value and its corresponding category, allowing users to easily understand the environmental conditions. A potentiometer is also connected to adjust the display contrast for better visibility.

2.2.7 Overall System Operation

The entire system operates as an integrated hardware solution, continuously monitoring air quality and updating the display in real time. The interconnection between components ensures smooth signal flow from sensing to processing and finally to output display. This design represents a complete embedded VLSI system that integrates sensor interfacing, communication protocols, digital processing, and user interaction on a single platform.

2.3 OPERATION: -

2.3.1 FPGA board (Zybo Z7-7000) -



Fig.4 FPGA board

Fig 5 shows, the Zybo Z7-7000 is a development board based on the Xilinx Zynq-7000 System-on-Chip (SoC), which integrates an ARM Cortex-A9 processor with programmable FPGA logic. This combination allows both hardware and software co-design, making it a powerful platform for embedded and VLSI applications. The board provides multiple input/output interfaces, including GPIO, SPI, UART, and other communication peripherals, along with on-board clock and memory support.

In this project, the programmable logic (FPGA) part of the Zybo board is used to implement digital modules such as the SPI controller, AQI calculator, AQI classifier, and LCD controller using Verilog HDL. The board operates at 3.3V logic levels, ensuring compatibility with various digital components. Its reconfigurable nature, high processing speed, and support for real-time operations make it well-suited for implementing the AQI monitoring system efficiently.

2.3.2 Smoke sensor –

MQ-135



Fig.5 MQ135 Smoke sensor

Fig 5 shows, the MQ135 gas sensor is widely used for detecting air quality and measuring the concentration of harmful gases such as carbon dioxide (CO_2), ammonia (NH_3), smoke, and other pollutants. It operates by sensing changes in resistance when exposed to different gas concentrations and produces a corresponding analog voltage output. This analog signal varies based on the level of air pollution in the environment.

In this project, the MQ135 sensor is used as the primary input device to monitor air quality. The analog output from the sensor is converted into digital form using an SPIbased ADC, enabling further processing by the FPGA. The sensor is cost-effective, easy to interface, and suitable for environmental monitoring applications, making it an ideal choice for the AQI monitoring system.

2.3.3 LED-



Fig.6 ADC MCP3008

Fig 6 shows, the MCP3008 is a 10-bit Analog-to-Digital Converter that uses the Serial Peripheral Interface (SPI) protocol for communication. It is capable of converting analog signals from multiple input channels into digital values, which can be processed by digital systems such as microcontrollers or FPGAs. The device operates at 3.3V, making it compatible with FPGA boards like the Zybo Z7-7000, and supports fast and reliable data transfer through SPI communication.

In this project, the MCP3008 ADC is used to convert the analog output of the MQ135 gas sensor into a digital signal. The FPGA communicates with the ADC using SPI signals such as SCLK, MOSI, MISO, and CS to read the converted data. This digital output is then used by the FPGA for AQI calculation and classification. The MCP3008 is chosen for its simplicity, multi-channel capability, and ease of interfacing with digital systems.

CHAPTER 3

3.1 Advantages: -

Real-Time Monitoring: The system continuously monitors air quality and provides instant AQI updates, allowing users to be aware of environmental conditions without delay.

Hardware Efficient: The use of FPGA enables fast and parallel data processing, reducing latency and improving system performance compared to software-based solutions.

Modern Communication: The integration of SPI protocol ensures efficient and reliable communication between the ADC and FPGA, making the design industryrelevant.

User-Friendly: The 16×2 LCD display provides clear and easy-to-understand information about AQI values and air quality status for users.

Scalable Design: The modular architecture allows easy expansion of the system to include multiple sensors and advanced features in the future.

Cost-Effective: The system uses readily available components, making it affordable for practical environmental monitoring applications.

3.2 Disadvantages: -

Sensor Accuracy Limitations: The MQ135 sensor provides approximate air quality readings and may require calibration for accurate and reliable results.

Environmental Dependency: Sensor readings can be affected by temperature, humidity, and other environmental factors, which may impact the accuracy of AQI calculation.

3.3 Applications: -

Environmental Monitoring: The system can be used to monitor air quality in homes, offices, and industrial areas, helping users stay aware of pollution levels.

Smart Homes: It can be integrated into smart home systems to automatically monitor and improve indoor air quality for better living conditions.

Industrial Safety: The system can be deployed in factories and industries to detect harmful gases and ensure worker safety.

Smart Cities: It can be used as part of smart city infrastructure for continuous monitoring of urban air pollution.

Public Health Awareness: The system helps in spreading awareness about air quality by displaying AQI levels in real time.

CHAPTER 4

4.1 Results: -

1. Simulation:-

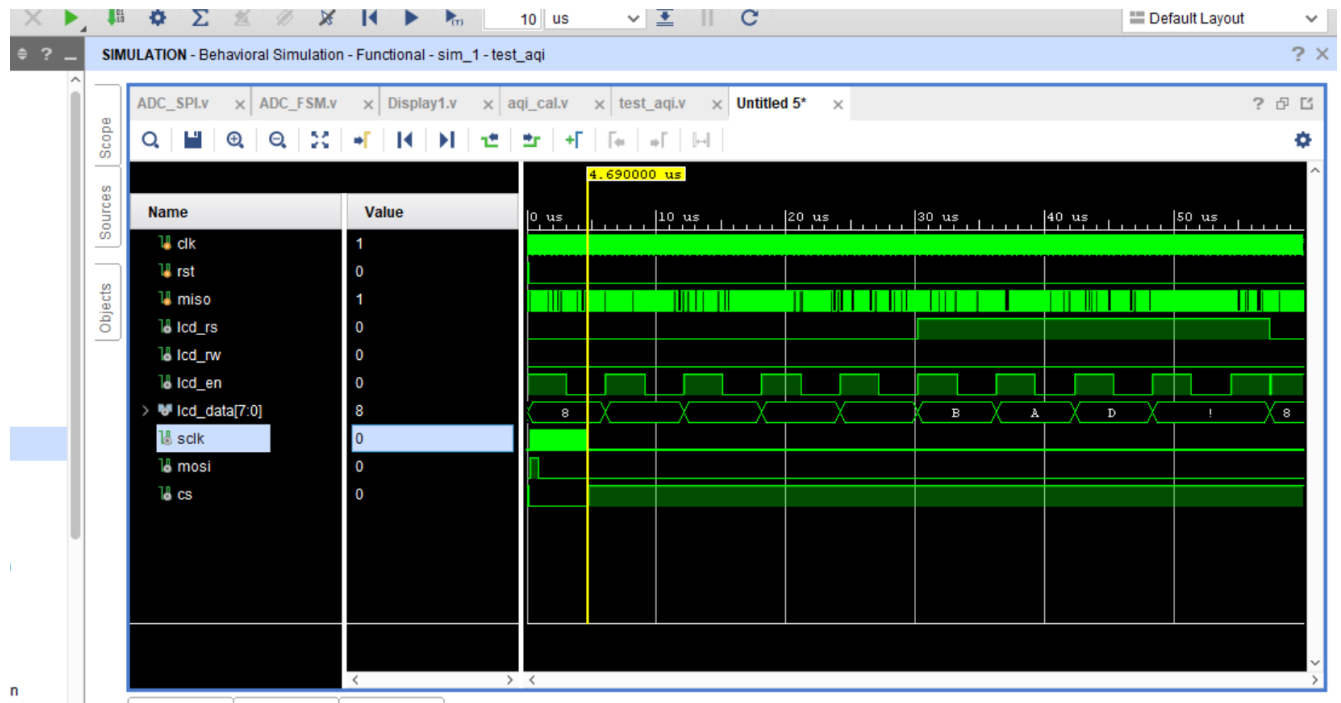


Fig.7 Simulation

Fig. 7 shows the simulation waveform of the SPI-based ADC interface integrated with the LCD display system implemented on the FPGA. The entire system operates using a 100 MHz clock, which acts as the reference for all sequential operations.

In the waveform, the Chip Select (CS) signal goes LOW to initiate communication with the ADC. Once enabled, the Serial Clock (SCLK) begins toggling, generating the required clock pulses for SPI data transfer. During this period, the MOSI (Master Out Slave In) line transmits the control command 11000, which corresponds to the start bit, single-ended mode, and selection of channel 0 in the ADC.

After receiving the command, the ADC sends the converted digital data through the MISO (Master In Slave Out) line. This data is captured serially and stored in a register within the FPGA. As shown in simulation we generated random patterns of MISO and as per the random values and there calculation it showing accurate output as BAD on lcd screen. The successful reception of data confirms proper SPI communication between the FPGA and ADC.

Subsequently, the LCD control signals, including RS (Register Select), RW (Read/Write), and EN (Enable), are activated. The LCD is first initialized using command instructions, after which the processed data is sent in data mode. The lcd_data bus in the waveform represents the characters being transmitted to the LCD, indicating correct display operation.

Thus, the waveform verifies that ADC communication, data processing, and LCD interfacing are functioning correctly, confirming the successful implementation of the system.

2. Schematic:-

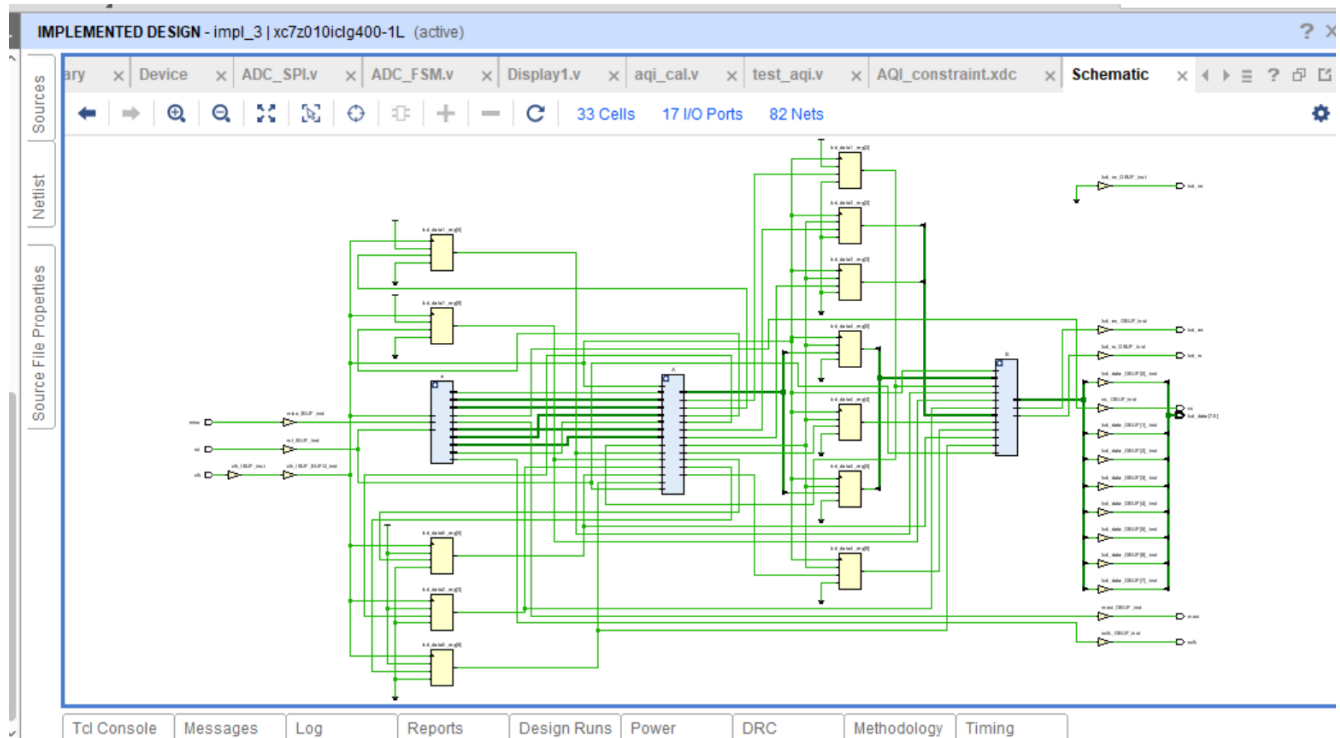


Fig.8 Schematic of AQI

3. Device Utilization: -

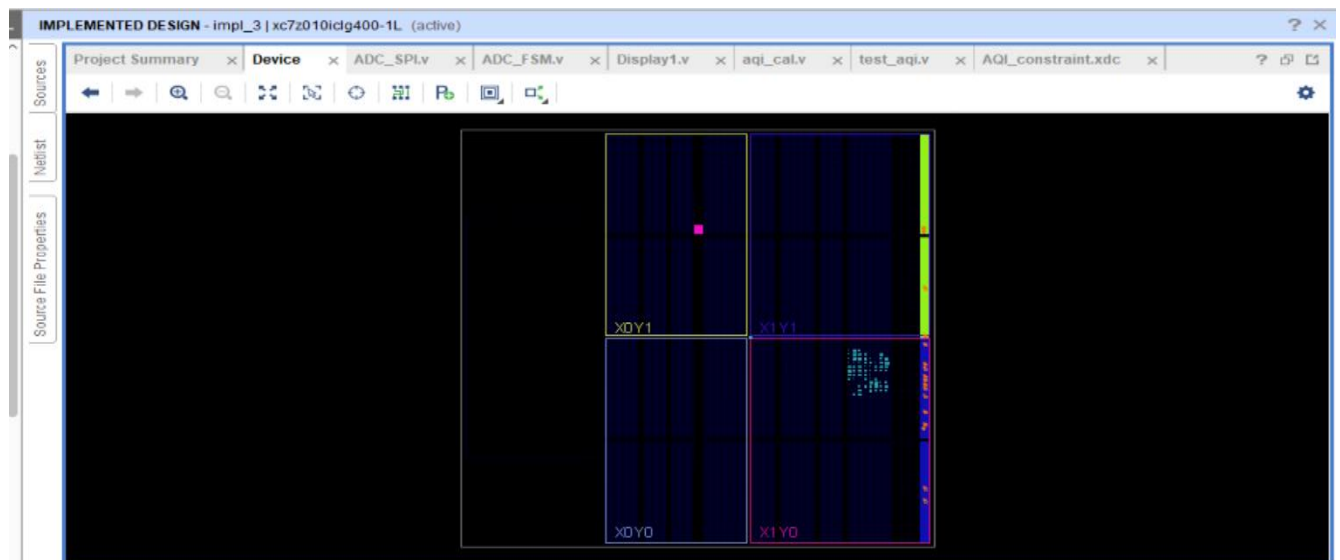


Fig. 9 FPGA Device Utilization and Placement

4. Power Analysis: -

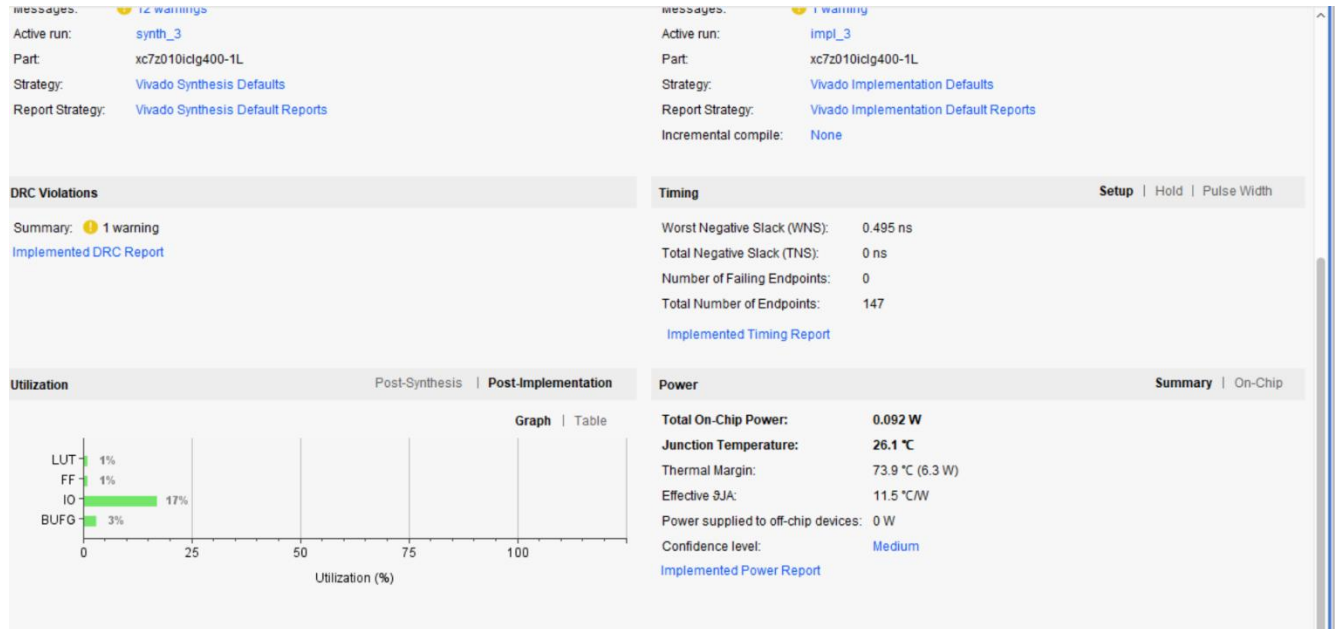


Fig.10 Implementation Results and Power Analysis

4. Operational Flow of Hardware Implementation: -

- **Sensor Monitoring:** The MQ135 gas sensor continuously monitors the surrounding air and detects the presence of harmful gases such as CO₂, ammonia, and smoke.

- **Analog Signal Generation:**

The sensor responds to air pollution by changing its internal resistance.

This change produces a corresponding analog voltage signal proportional to the pollution level.

- **Analog-to-Digital Conversion:**

The analog signal from the sensor is sent to the MCP3008 ADC.

The ADC converts this analog signal into a digital value using SPI communication.

- **Data Acquisition (SPI):**

The FPGA communicates with the ADC using SPI signals (SCLK, MOSI, MISO, CS).

The ADC sends the converted digital data to the FPGA.

- **AQI Calculation:**

The FPGA receives the digital data and processes it using Verilog logic. The data is converted into a normalized AQI value.

- **AQI Classification:**

The calculated AQI value is compared with predefined ranges.

The system classifies air quality into categories such as Good, Moderate, or Poor.

- **Display Output:**

The FPGA sends the AQI value and status to the 16×2 LCD.

The LCD displays the information in a clear and user-friendly format.

- **Continuous Monitoring:**

The system continuously repeats the process to provide real-time air quality updates.

- **Repeat Operation:**

As air quality changes, the system updates the AQI value and display accordingly without user intervention.

4.2 Conclusion: -

This project successfully demonstrates the design and implementation of a VLSI-based Air Quality Index (AQI) monitoring system using FPGA. The system effectively acquires air quality data from the MQ135 sensor, converts it into digital form using an SPI-based ADC, and processes it using Verilog-based logic. The calculated AQI value is classified into standard air quality categories and displayed on an LCD, providing realtime and user-friendly output.

The use of FPGA enables fast and efficient hardware processing, while the integration of SPI communication makes the system modern and scalable. Overall, the project highlights the feasibility of implementing environmental monitoring systems using VLSI design techniques and provides a strong foundation for future enhancements such as multi-sensor integration and System-on-Chip (SoC) development.

References: -

- 1) Central Pollution Control Board (CPCB), *National Air Quality Index*, Available at: <https://cpcb.nic.in/National-Air-Quality-Index/>
- 2) Air Quality Index (AQI) Basics, Available at: <https://www.airnow.gov/aqi/aqi-basics/>
- 3) UiTM Institutional Repository, Research Paper on Air Quality Monitoring System, Available at: <https://ir.uitm.edu.my/id/eprint/124087/1/124087.pdf>
- 4) Digilent Inc., “Zybo Z7 FPGA Board Reference Manual and Master XDC File,” [Online]. Available: <https://digilent.com/reference/programmable-logic/zybo-z7/start>

Appendix: -

A) CODE: -

1) FSM

```
module AQI_FSM(  
    input clk,rst,miso,  
    output lcd_rs,  
    output lcd_rw,  
    output lcd_en,  
    output [7:0] lcd_data,  
    output sclk,  
    output cs,  
    output mosi  
);  
    wire aqi_done;  
    wire [7:0] adc_val;  
    wire [7:0] aqi_val;  
    reg [7:0] lcd_data1;  
    reg [7:0] lcd_data2;  
    reg [7:0] lcd_data3;  
    reg [7:0] lcd_data4;  
    reg [7:0] lcd_data5;  
    ADC_SPI a(.clk(clk)  
        ,.rst(rst)  
        ,.miso(miso)
```

```
    ,.adc_val(adc_val)
    ,.sclk(sclk)
    ,.cs(cs)
    ,.mosi(mosi)
);
```

aqi_cal

```
A(.AQI_VAL(aqi_val),.clk(clk),.rst(rst),.adc_val(adc_val),.aqi_done(aqi_done));
Display1 B(.clk(clk),.rst(rst),.lcd_rs(lcd_rs),.lcd_rw(lcd_rw),
    .lcd_en(lcd_en),.lcd_data(lcd_data),.lcd_data1(lcd_data1),
    .lcd_data2(lcd_data2),.lcd_data3(lcd_data3),
    .lcd_data4(lcd_data4),.lcd_data5(lcd_data5)
);
```

```
always @(posedge clk) begin
```

```
    if (rst) begin
```

```
        lcd_data1 <=8'h0; end
```

```
    /* else if(aqi_done == 1'd1) begin
```

```
        aqi <= aqi_val;
```

```
        adc<= adc_val;*/
```

```
    if (aqi_val >= 8'd0 && aqi_val <= 8'd67) begin
```

```
        lcd_data1 <="G";
```

```
        lcd_data2 <="O";
```

```
        lcd_data3 <="O";
```

```

        lcd_data4 <="D";
        lcd_data5 <="!";

    end

else if (aqi_val >= 8'd68 && aqi_val <= 8'd136) begin
    lcd_data1 <="B";
    lcd_data2 <="A";
    lcd_data3 <="D";
    lcd_data4 <="!";
    lcd_data5 <="!";

    end

else if(aqi_val >= 8'd137 && aqi_val <= 8'd200) begin
    lcd_data1 <="w";
    lcd_data2 <="O";
    lcd_data3 <="R";
    lcd_data4 <="S";
    lcd_data5 <="T";

    end

    end

//end

endmodule

```

2) ADC

```
`timescale 1ns / 1ps
module ADC_SPI(
    input clk,
    input rst,
    input miso,
    output reg [7:0] adc_val=8'd0,
    output reg sclk=0,
    output reg mosi=0,
    output reg cs=1

);

reg [1:0] state =0;
reg [4:0] bit_count =5'd0;
reg [7:0] shift_reg = 8'd0;
reg [3:0] counter = 4'd0;
reg [4:0] cmd =5'b11000; //logic for ch0

always @(posedge clk or posedge rst) begin
    if(rst) begin
        state <= 0;
        cs <= 1;
```

```

mosi <= 0;
shift_reg<=8'd0;
bit_count <= 5'd0;
counter <= 4'd0;
end
else begin
case(state)

2'd0: begin
cs <= 0;

bit_count <= 5'd0;
shift_reg<=8'd0;
counter <= 4'd0;
sclk<=1'd0;
state <= 2'd1;

end

2'd1: begin
counter <= counter +1'd1;
if(counter == 4'd8) begin
counter <= 4'd0;           // serial clk generator
sclk <= ~sclk;

```

```

if(sclk == 1'd0) begin

    // ? SEND MOSI FIRST (for first 5 bits)
    if(bit_count < 5)
        mosi <= cmd[4 - bit_count]; // for ch0 of adc
    else
        mosi <= 0;
end

else begin
    shift_reg <= {shift_reg[6:0], miso};

    if(bit_count == 5'd12 )
        state <= 2'd2;
    else
        bit_count <= bit_count + 1;
    end
end

end

end

2'd2: begin
    cs <= 1;
    sclk <= 0;
    adc_val <= shift_reg;

```

```

        //state <= 2'd0;
    end
    default : begin    cs <= 0;

        bit_count <= 4'd0;
        shift_reg<=8'd0;
        // state <= 2'd1;
        counter <= 4'd0;
        sclk<=1'd0; end
    endcase
end

end

endmodule

```

3) AQI CALCULATOR

```

module aqi_cal(
    input clk,rst,
    input [7:0] adc_val,
    output reg [7:0] AQI_VAL,
    output reg aqi_done
);

```

```

parameter AQI_MAX = 8'd200;
parameter ADC_MAX = 8'd255;
reg [15:0] mul_result;

always @(posedge clk ) begin
if(rst) begin
AQI_VAL <= 8'd0;
aqi_done<=1'd0;
mul_result <= 16'd0;
end
else begin
//if(adc_done == 1'd1)

    mul_result <= adc_val * AQI_MAX;
    AQI_VAL <= mul_result / ADC_MAX;           // aqi calculator
    //AQI_VAL <= adc_val * 0.7843;
    aqi_done <=1'd1;
end
end

endmodule

```


4) LCD Display

```
`timescale 1ns / 1ps
```

```
module Display1(
```

```
    input clk,rst,
```

```
    output reg lcd_rs =1'b0,
```

```
    output reg lcd_rw =1'd0,
```

```
    output reg lcd_en =1'b1,
```

```
    output reg [7:0] lcd_data = 8'd0,
```

```
    input [7:0] lcd_data1,
```

```
    input [7:0] lcd_data2,
```

```
    input [7:0] lcd_data3,
```

```
    input [7:0] lcd_data4,
```

```
    input [7:0] lcd_data5
```

```
);
```

```
    reg flag;
```

```
    reg [16:0] count = 17'd0;
```

```
    reg [4:0] state = 5'd0;
```

```
always @(posedge clk) begin
```

```
if (rst) begin
```

```
    count <= 4'd0;
```

```
    flag <= 1'b0;
```

```
end
```

```
else if (count == 17'd100000) begin    // clock divider for lcd 1ms delay
```

```
    count <= 4'd0;
```

```
    flag <= 1'b1;
```

```
end
```

```
else begin
```

```
    count <= count + 1'b1;
```

```
    flag <= 1'b0;
```

```
end
```

```
end
```

```
always @(posedge clk) begin
```

```
if (rst || state == 5'd19 ) begin // state rst
```

```
    state <= 0; end
```

```
else if (flag) begin
```

```

        state <= state + 1'd1;

    end

    else begin state <= state ; end

case (state)           // lcd execution steps

5'd0: begin

    lcd_rs <= 1'd0; lcd_en <= 1'd1; lcd_rw <= 1'd0; lcd_data<= 8'h38;

    end

5'd1: begin

    lcd_rs <= 1'd0; lcd_en <= 1'd0; lcd_rw <= 1'd0;

    end

5'd2: begin

    lcd_rs <= 1'd0; lcd_en <= 1'd1; lcd_rw <= 1'd0; lcd_data<= 8'h0c;

    end

5'd3: begin

    lcd_rs <= 1'd0; lcd_en <= 1'd0; lcd_rw <= 1'd0;

    end

5'd4: begin

    lcd_rs <= 1'd0; lcd_en <= 1'd1; lcd_rw <= 1'd0; lcd_data<= 8'h06;

    end

```

5'd5: begin

 lcd_rs <= 1'd0; lcd_en <= 1'd0; lcd_rw <= 1'd0;

 end

5'd6: begin

 lcd_rs <= 1'd0; lcd_en <= 1'd1; lcd_rw <= 1'd0; lcd_data <= 8'h01;

 end

5'd7: begin

 lcd_rs <= 1'd0; lcd_en <= 1'd0; lcd_rw <= 1'd0;

 end

5'd8: begin

 lcd_rs <= 1'd0; lcd_en <= 1'd1; lcd_rw <= 1'd0; lcd_data <= 8'h80;

 end

5'd9: begin

 lcd_rs <= 1'd0; lcd_en <= 1'd0; lcd_rw <= 1'd0; end

5'd10: begin lcd_rs <= 1'd1; lcd_data <= lcd_data1; lcd_en <= 1'd1; end

 5'd11: begin lcd_rs <= 1'd1; lcd_en <= 1'd0; end

 5'd12: begin lcd_rs <= 1'd1; lcd_data <= lcd_data2; lcd_en <= 1'd1; end

 5'd13: begin lcd_rs <= 1'd1; lcd_en <= 1'd0; end

 5'd14: begin lcd_rs <= 1'd1; lcd_data <= lcd_data3; lcd_en <= 1'd1; end

 5'd15: begin lcd_rs <= 1'd1; lcd_en <= 1'd0; end

```
5'd16: begin lcd_rs <= 1'd1; lcd_data <= lcd_data4; lcd_en <= 1'd1; end

5'd17: begin lcd_rs <= 1'd1; lcd_en <= 1'd0; end

5'd18: begin lcd_rs <= 1'd1; lcd_data <= lcd_data5; lcd_en <= 1'd1; end

5'd19: begin lcd_rs <= 1'd1; lcd_en <= 1'd0; end

default : begin lcd_rs <= 1'd0; lcd_en <= 1'd1; lcd_rw <= 1'd0; lcd_data <=
8'h38; end

endcase

end

endmodule
```

